

The Axis of the Sign Is Not Scale: Evaluation Protocol Determines the Sign of the Scale Effect in Multimodal RAG

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Abstract

When retrieved evidence conflicts with a model’s parametric knowledge, behaviour is no longer governed by task difficulty alone. Holding the conflict construction, the adjudication criterion, the models and the samples fixed, we vary exactly one variable: the strength with which the system prompt authorises parametric knowledge, pre-registered as seven ordered stances from weak to strong. On 927 controlled counterfactually edited conflict samples, an anchoring protocol guarantees that every scale point genuinely knows the fact at issue ($|D^*| = 649$). The result is that the sign of the scale effect is determined by the evaluation protocol, not by scale. The Δ of the seven stances crosses zero, with signs running from negative to positive along the pre-registered authorisation order, spanning 47.5 to 56.5 percentage points, and replicating independently on two model families. The sign flip is decided by the mid-ladder stance `ctx_hedge`, so the ordering is not a monotone gradient, and the structure holds only under free generation. Methodologically we report three directional measurement biases—explicit rejection misread as undecidable, denominator collapse misread as the strongest effect, and position preference masquerading as content judgment—each of which is sufficient to let the same data support the opposite conclusion.

Keywords: knowledge conflict, multimodal RAG, evaluation protocol, scale effect, sign crossing, measurement bias

1 Introduction

Retrieval-augmented generation delegates facts the parameters cannot hold to external evidence. Once retrieved content conflicts with parametric knowledge, behaviour is no longer determined by task difficulty: for the same question and model, the answer may follow retrieved content or the parameter, and which one is chosen depends on how the prompt is written. This is not a corner case—retrieved evidence may itself be wrong, and whether the model holds to parametric knowledge decides whether the system keeps that error out or accepts it wholesale. Multimodal settings push the problem further: image evidence, retrieved text, and parametric facts may conflict pairwise.

A recent body of work measures this choice: ConflictBank constructs samples in which context and parametric knowledge conflict [1], ClashEval reports adoption of incorrect retrieved content above 60% [2], and the multimodal sycophancy benchmark PENDULUM extends the measurement to vision-language models [3]. These establish that models are swayed by conflicting context and leave the next question: does this swaying improve or worsen as capability increases?

The natural guess is that it improves—parametric knowledge grows with scale, so models should better identify incorrect evidence. But testing this carries an easily overlooked precondition: scale must be the only variable that changes. In practice the prompt changes too, some saying "answer only from the context", others "if the context is unreliable, rely on your own knowledge". If the degree to which the template authorises parametric knowledge itself changes behaviour, attributing a difference between scales to scale may amount to measuring the template.

This paper turns the question into a decidable experiment: fixing the conflict construction, the decision criterion, the models, and the samples, and varying a single variable—the strength with which the system prompt authorises parametric knowledge, pre-registered by semantics from weak to strong into seven levels, from "use only the context" to "rely on your own knowledge". On 927 conflict samples from controlled counterfactual edits we measure the proportion of cases where the model follows parametric knowledge at each scale and stance. An anchoring protocol ensures every scale under test in fact knows the fact—if a small model does not know it, its "not following parametric knowledge" under conflict is not a product of choice. This shrinks 927 samples to 649 and isolates the capability gap from the trivial explanation that larger models simply know a few more facts.

The result is that the sign of the scale effect is determined not by scale but by stance. Across two model families, weakly authorising stances make larger models more inclined to follow the context (**ctx_only** at -27.3 and -8.4 pp), while strongly authorising stances make them likelier to hold to parametric knowledge (**prior** at $+28.0$ and $+33.7$ pp). The Δ of both families crosses zero, the sign is ordered from negative to positive along the pre-registered authorisation order, and decomposition into adjacent scale pairs shows early saturation rather than smooth growth.

That the signs are ordered is not a monotonic gradient, and this is the paper's most critical and most easily misread point. What decides the flip is not the rule "the stronger the authorisation, the more positive the effect": the mid-sequence stance **ctx_hedge** explicitly commands "when the context is unreliable, rely on your own knowledge" and yields $+29.2$ pp for the Qwen3 family, whereas the semantically weaker

neutral stance that follows yields only +1.5 pp. Which stance the sign lands on is decided jointly by that stance’s output form and the decision criterion—the “protocol” rather than the “stance semantics” of the title. This cell is also the least trustworthy in magnitude, yet removing it entirely leaves the sign sequence of both families unchanged (first two negative, last four positive), the span falls only from 56.5 to 55.3 pp, and the Qwen3 rank correlation rises from 0.643 to 1.000. The least trustworthy cell is not load-bearing.

The conclusion carries a necessary qualification that is itself part of the result: it holds only under the free-generation format. Under the forced-choice template of two published works, the scale effects of all four conditions stay within -4 pp, uniformly negative, and do not change with the framing (4.3).

At the methodological level we report three measurement problems not previously addressed systematically, each of which would lead the same data to conclusions of opposite direction. First, the criterion misrecords explicit rejection as undecidable: when a model outputs “the context is incorrect, the origin of X is Y” it names both values and is judged undecidable and excluded, yet such answers are the most unambiguous cases of following parametric knowledge; exclusions rise with scale, and after correction the PKD of one stance rises from 0.359 to 0.960. Second, denominator collapse is misread as the strongest effect: paired decidable samples for the same stance collapse from 584 to 1 across three criteria, and when the denominator reaches single digits Δ can only take degenerate values, yet in the table it looks like the strongest cell; we therefore stipulate that cells with $n < 50$ are not counted in the span and sign comparisons. Third, position preference can completely override content: under the forced-choice format, rerunning after swapping the option contents yields a position preference of +0.961, so the high accuracy is a product of position rather than content judgement. None of the three disappears when the criterion or template is changed, and each has at some point reversed the direction of one of our own conclusions.

The contributions are three. First, the measurement conclusion that the sign of the scale effect is a function of the evaluation protocol, turned into a decidable design: seven pre-registered stances $\times$ five scale points $\times$ two model families, with Δ crossing zero and the signs ordered, independently reproduced across the two families. Second, the three-way conflict construction of controlled counterfactual edits plus an anchoring protocol, separating “whether the model resists tampered retrieved evidence” from “whether the model knows the fact”, so that cross-scale comparison has determinate semantics. Third, three directionally clear measurement biases, each reported with its direction, magnitude, and before-and-after comparison.

We state the boundaries honestly. Measurement covers only five scale points, two model families, and a single dataset, all under text conditions (images provide entity recognition, answers are given as text). The mechanistic explanation is behavioural: we show the sign varies with stance, but do not claim to have located the internal representational mechanism. Work discovering the negative and the positive directions each has predecessors, and “the effect sign can cross zero” has also been reported along the model-size axis [7]; we claim no priority for these, but claim the orthogonality conclusion that the axis crossing zero is the evaluation protocol rather than model

size, and the separation evidence that under length control the gradient is driven by authorisation semantics rather than answer length.

Owing to the length limit, the supporting tables and the code are available in the online material at <https://github.com/data-echo-20/slim-eval-protocol>.

2 Related Work

2.1 Conflict between parametric knowledge and context

How a model chooses when context disagrees with its parametric knowledge is the starting point of knowledge-conflict research. ConflictBank [1] constructs samples in which the two conflict (7.45M claim–evidence pairs, 553K question–answer pairs); Longpre et al. [2] formalise conflict as an entity-level “context-versus-parametric” conflict and argue that over-reliance on memorised information causes hallucination; ClashEval [4] applies graded perturbations to over 1,200 questions across 6 domains and reports adoption of erroneous retrieved content above 60%. Mechanistically, Yao et al. [18] and Pham et al. [20] show via knowledge circuits and layer-wise attention attribution that the competition occurs at locatable intermediate layers, not as output-layer weighting. All of these measure how often a model is “led astray”; we measure the sign direction and cross-stance ordering, and introduce a three-way conflict structure—a counterfactual edit supplying tampered retrieved text, image evidence, and parametric knowledge at once.

2.2 Scale as an independent variable

Scale and capability are usually related by power laws—Kaplan et al. [5] find loss decreasing as a power law with model size, data, and compute, whereas McKenzie et al. [6] report U-shaped and inverted-U non-monotonic trends. Most relevant is the interaction of scale with instruction following. De Marez et al. [7], across 56 open-weight models from six families spanning 0.3B–32B, use two-alternative prompts to measure factual sycophancy under social pressure and find that the advantage of instruction-tuned over base models reverses sign with scale, crossing zero near 7B. That work is both our closest competitor and our most powerful control: it crosses zero along the model-size axis while holding the protocol fixed, whereas we cross zero along the protocol axis while holding the family and samples fixed. Its 13 manipulation classes all lie within the social-pressure framing and its dependent variable is the log-probability difference between two fixed alternatives, so abstention does not exist structurally; it contains no inference-time manipulation of authorisation strength, no dose–response, and no length control. Since it already reports a zero crossing at 7B in the same statistical language, we claim no priority for showing an effect sign can cross zero, only for swapping the axis along which the crossing occurs. Its explanation of the crossing—the threshold artefact, where removing the mean difference between the two channels drops the interaction variance share by about 79.9%—must be addressed head-on; we counter with three moves it structurally cannot make: reusing the same question set across protocols, an item-level sign test rather than a comparison of aggregate means, and abstention as a second independent curve. Scale and fine-tuning

are further entangled: Zhou et al. [21] train models to trade off between parametric knowledge and context, Gekhman et al. [29] give counter-evidence (knowledge injected by fine-tuning correlates positively with hallucination rate), and Wei et al. [30] show task format alone can elicit zero-shot ability; we vary authorisation, not format.

2.3 The effect of the evaluation protocol on conclusions

Our position rests on a general observation: the same batch of outputs can be judged as different behaviours under different protocols. Sycophancy and instruction-following measurements are wording-sensitive—Sharma et al. [8] attribute factual sycophancy to human preference judgments favouring sycophantic replies, and Turpin et al. [9] show bias features such as option reordering can shift answers by as much as 36% without being mentioned. On the retrieval side, Asai et al. [24], Wang et al. [22], and Huang et al. [23] eliminate conflict through self-assessed retrieval, iterative evidence filtering, and conditional trust; we do not eliminate it but treat it as a measurement instrument. Sycophancy also has training-side sources: synthetic data can suppress it without costing general capability [41], and once extended from factual to social it connects to reward tampering [42, 43]. BehnamGhader et al. [25] attribute reasoning failures under retrieval augmentation to the division of labour between retriever and language model, a different facet of the same interference we observe. Two works are closest and must be disentangled. Pandere et al. [10] perform circuit-level attribution on base and instruct models, five methods consistently pointing to gating rather than rewiring, and characterise instruction tuning as pushing the model toward parametric memory; the paper lists model scale as a limitation and notes the effect disappears for coherent evidence passages, a framing factor. Bhowal et al. [15] distinguish two operationalisations of "the model's own reasoning" in medical vision-language models—edited text occupying the model's own token positions (prefix-forced continuation) versus merely being source-tagged in the prompt—where the same text is adopted about 73.5% of the time under the former and 16.0% under the latter. Its argument runs opposite to and complements ours: it treats direction reproduction as robustness evidence and attributes the magnitude gap to model specificity, whereas we argue the sign flips across zero with protocol strength, so "direction unchanged, magnitude variable" is a counterexample rather than a precedent. Their protocol axes are binary (restated prompt versus prefix forcing, trust instruction on/off), whereas our authorisation strength is a seven-level pre-registered ordinal variable; and neither conducts a zero-crossing test.

2.4 Correspondence under multimodal conditions

Multimodal conditions push the conflict to visual evidence. Liu et al. [11] examine 9 multimodal large models on 374 images and 1,122 question-answer pairs, reporting that on about 20% of queries the model over-relies on parametric knowledge rather than the contradicting visual evidence; CC-VQA [12] is a training-free mitigation achieving 3.3–6.4 pp gains on E-VQA, InfoSeek, and OK-VQA. Tang et al. [13] divide conflicts in multimodal long-chain reasoning into goal conflicts at the input level and effective conflicts at the process level, reporting directional asymmetry; Lee et al. [14] find across 13 multimodal large models that the same semantics yields different

results as an image versus as text, and that models accept contradictory context more easily in image form. Li et al. [34] measure object hallucination with polling-style discriminative queries, separating "misseeing" from "mismemorising". PENDULUM [3] extends sycophancy measurement to vision-language models with the interlocutor's prompt influence direction as its independent variable, whereas ours is the authorisation strength for parametric knowledge. Aranya et al. [16] oppose "grounding" to "sycophancy" and report the two negatively correlated, but fix all models at the same order of magnitude and use only non-negative dependent variables, so they occupy no scale dimension and structurally cannot cross zero.

3 Data and measurement protocol

3.1 Dataset and filtering

3.1.1 From E-VQA to two generations of anchoring sets

Our evaluation set is built on E-VQA, whose questions refer to their entity through a referring phrase ("this plant", "this animal"), so the entity's identity can be determined only from the image. That is an advantage in the original task but an obstacle for our quantity: feed the referring expression verbatim and what the model answers is the linguistic **prior** of that question type, not knowledge about the species. The symptom surfaced in the first round of filtering: the 6 samples flagged "strongly anchored" all fall on one template — 'In which season does this plant give flowers? **is always answered** Spring' across 6 different species. What the model gets right is the common-sense distribution of flowering time, independent of whether it recognises the species. The remedy was to replace the referring expression with the species name and re-run the filtering, which lowers the hit rate but buys a reliable measurement of "the model holds this fact" rather than "the model is familiar with this sentence pattern". Recent multimodal knowledge-conflict benchmarks likewise centre the question on the entity [31], the conflict lying in the inconsistency between the model's parametric memory of that entity and the context. The two generations of anchoring sets, whose sizes and filtering criteria are in Table S11 (online material), barely overlap: of the 927, only 147 belong to the 236. The first generation got the editing pipeline working and the second is the object of study here; keeping them apart matters because the first was filtered by a single model and carries a selection bias — facts known by the 3B model skew towards high-frequency, templated species attributes, exactly the ones a large model can answer without retrieval, so comparing scales on such a set absorbs the difference into the filtering criterion.

3.1.2 Construction of the second-generation set

The second-generation set is built in three steps: take the entries with evidence sentences from the E-VQA candidate pool, re-ask them with the entity-name phrasing, have Qwen2.5-VL 3B and 7B each produce one context-free answer, and keep the samples for which both produce the parametric ground truth. Two scales rather than one are used because our dependent variable is the behavioural difference across scales: filtering with a single model would select samples necessarily "known" to it while leaving

untested whether the other scale knows them too, and only when both are confirmed to know them is any later difference guaranteed not to be the trivial effect of "a larger model knows a few more facts". Each sample then undergoes counterfactual editing; 1146 enter the valid pool, of which 927 pass quality control and constitute the main set. The other 219 fail rewriting — the edited text is semantically self-contradictory, the number format is malformed, or the edit does not actually change the value — and have an empty `context_conflict` field; at evaluation time they are fed a context consistent with parametric knowledge, so they can never exhibit conflict, and leaving them in the denominator dilutes PKD with zeros.

3.1.3 Counterfactual editing pipeline

The carrier of the edit is a single slot in the evidence sentence; after editing, the context conflicts with the parametric ground truth while everything else stays verbatim. Single-slot substitution is necessary: it makes the conflict location precisely identifiable, so that "did the model notice the conflict" and "on which semantic dimension did the model concede" can be examined separately. Rewriting a whole passage cannot do this. Conflict injection is cognate with existing knowledge-conflict benchmarks [17, 19]; the only difference is granularity — baseline work usually injects a whole conflicting evidence passage, whereas we touch a single slot.

The pipeline handled four classes of failure. The disposition of the four classes is listed in Table S1 (online material).

The final quality-control result is malformed numbers 0, duplicated punctuation 0, edits that failed to take effect 0.

3.1.4 Image mapping

All 236 anchoring samples are matched with iNaturalist 2021 images; the mapping chain has three segments:

```
wikipedia.title --(scientific name match)--> val.json.categories.name
-> categories.image_dir_name -> images.file_name
```

The scientific-name match rate of the first segment is measured at 100%. Two error-prone points are recorded below. First, `dataset_category_id` must be zero-padded to 5 digits, or the directory-name match fails. Second, `dataset_image_ids` stores the original iNat photo numbers, which are not among the 100,000 published for the competition and cannot be used to locate files; one must go through the `image_dir_name` and `file_name` fields.

3.1.5 Slot composition

The coarse-grained slot distribution of the 927-sample main set is place 512, other 335, quantity 33, diet 28, time 19; place accounts for 55.2%, so the conclusions rest mainly on geographical facts. At the fine-grained level the top `edit_slot` categories are place 507, city 281, quantity 44. One reading caution: the quantity slot has only 33 entries here against 66 in the first-generation 236 set, its share dropping from 28.0% to 3.6%, so the two generations' slot compositions differ considerably and we draw no per-slot

$$\text{Construction and cost of the anchoring protocol } \mathcal{D}^* = \bigcap_s \{i : K_s(i) = 1\}$$

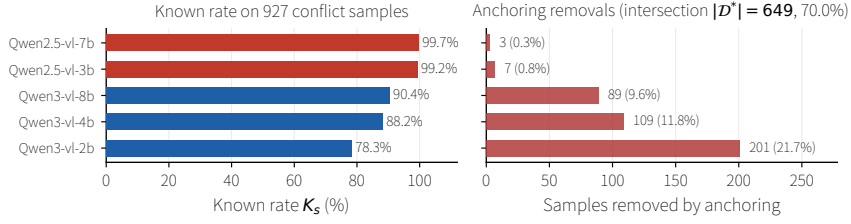


Fig. 1 Construction and cost of the anchoring protocol $\mathcal{D}^* = \bigcap_s \{i : K_s(i) = 1\}$.

scale-effect conclusion, reporting it only alongside the domain-wise tests of 4.5 with each group’s sample size. The scale comparison is performed not on these 927 but on their subset $\mathcal{D}^*$ (4.1.5).

Figure 1. The left panel gives the rate K_s at which each scale judges a conflict sample as known on the common-coverage set; the right panel gives the number of samples removed by anchoring. The anchored set keeps only the samples every scale judges known; the price is a shrunken denominator, the gain is the removal of the background condition that the amount of knowledge varies with scale.

3.2 Measurement protocol and implementation details

Every conclusion in this chapter rests on one kind of measurement: give the model a context that conflicts with its own knowledge and see whom it follows. Such measurements are unusually sensitive to implementation details, and our pilot experiments found three choices that make the same batch of samples yield completely different results. We state them before the results because every number that follows depends on them, and because these errors are widespread in published work: a reader who does not reproduce our protocol choices cannot obtain comparable numbers.

3.2.1 Three measurement conventions

Convention one: the context-free arm must use a neutral system prompt

Measuring “how much the model knows on its own” requires withholding the context, but the system prompt still affects the measurement: reusing the context arm’s prompt tells the model, under a condition where it should rely on its own knowledge, “not to rely too much on yourself”, and on 236 samples the parametric knowledge rate is 0.496 with that prompt against 0.9746 with a **neutral** one — nearly 48 pp, enough to reverse the basic judgement of “whether the model knows”. The context-free arm therefore uses a separate **neutral** prompt, **Youareaknowledgeassistant.Answerwithashort phrase.**; the seven stances of the context arm each have their own wording (3.2.2).

Convention two: the knowledge rate is read out only on the text-only arm

The image arm measures "how much parametric knowledge remains after the image intervenes", which is not an unbiased estimate of knowledge held; filtering samples with it would mix visual interference into the selection. Our anchoring set D^* is therefore constructed entirely from the text-only arm; the visual arm is used only for the comparison in 4.4.

Convention three: the effective denominator of the conflict condition must exclude samples whose rewriting failed

Convention three is as in 3.1.2: a sample whose rewriting failed has a context consistent with parametric knowledge and can never exhibit conflict, so counting it would dilute PKD with zeros; the effective denominator is 927 after the 219 failed rewrites are removed.

3.2.2 The seven instruction stances

The seven stances change only the wording of the system and user templates; the samples, the context, the image, and the decoding parameters are completely unchanged. Table 1 gives the verbatim definitions and the authorisation strength. Authorisation strength is the degree of authorisation of parametric knowledge, pre-registered from weak to strong; the ordering is fixed before the data are seen.

Table 1 Verbatim definitions of the seven instruction stances and their authorisation strength

Stance	system prompt	user template
ctx_only	...thatreliessolelyontheprovidedcontext.	Answerbrieflyusingonlythecontext:
orig	...usingtheprovidedcontextwhenrelevant.	Answerbrieflybasedonthecontextandtheimage:
ctx_hedge	...ifyoubelievethethecontextisinaccurate,answerfromyourownknowledge.	Answerbriefly,correctingthecontextifit iswrong:
neutral	Youareaquestionansweringsystem.	Answerbriefly:
own_only	...thatreliessolelyonyourownknowledge.	Answerbrieflyusingonlyyourownknowledge:
len_ctrl	Usetheprovidedcontextandyourownknowledge together, and answer as appropriate.	Answerbrieflyusingthecontextandyourown knowledge as appropriate:
prior	Use your own knowledge when it is more reliable than the provided context.	Answerbriefly, preferring your own knowledge if it is more reliable:

Three stances need their rationale explained. **orig** copies the reproduced work’s prompt verbatim, including **andtheimage**; the wording looks superfluous on the text-only arm, but changing it would mean it is no longer a reproduction, and keeping it in the same run rather than taking numbers from another run eliminates batch differences — the seven rows of the main table come from one decoding run, with the same server-side state and batch effects. **ctx_hedge** covers the intermediate state real retrieval systems use: the first four stances sit between "instruct to use the context" and "permit use of one’s own knowledge", and a natural objection is that the gradient is manufactured by pushing to the extremes, so **ctx_hedge** tests the intermediate state directly and provides the gradient’s zero-crossing point. **len_ctrl** is the length-control

placebo: the first four stances differ in word count (`ctx_only` 13, `neutral` 7, `prior` 17, `own_only` 14), so the gradient covaries with length, whereas `len_ctrl` equals `prior` in length and is isomorphic in structure (17 words, ‘Use X when Y’) but **semantically neutral, with** appropriate granting precedence to neither side. Had it fallen near `neutral` rather than `prior`, the gradient would come from semantics rather than length; measured empirically it falls between the two (4.1.3).

3.2.3 Decoding and evaluation parameters

The decoding and evaluation parameters are in Table S5 (online material).

Temperature is fixed at zero for reproducibility, at the cost of being unable to report the variance across runs. We substitute paired testing: the same batch of samples receives the seven stances within the same run, the comparisons between stances are a paired design, and the model’s own sampling noise cancels in the paired differences. Zero temperature does not guarantee deterministic output; the non-determinism introduced by inference frameworks and batching has been documented empirically [38, 39]; the paired design is our response to this, but it cancels noise within a run, not drift between runs. The concurrency settings and the service-readiness probe are in the reproduction material.

3.2.4 Quantisation: one confound that must be disclosed

All main experiments use bf16 precision. The first-generation results did not: 3B used fp16 and 7B AWQ quantisation, a mixed-precision comparison whose “parametric knowledge following rate decreases with scale” was a quantisation artefact rather than a scale effect. Quantisation damages precisely the long-tail fact recall this research measures: the same Qwen2.5-VL-3B has a parametric knowledge rate of 0.9746 under bf16 but 0.6864 under AWQ (a loss of 28.8 pp), and the unquantised 3B knows far more than the quantised 7B (0.9746 versus 0.6017, 37.3 pp). Our second methodological conclusion is therefore that cross-precision scale comparisons are unusable here. This is also why we abandoned Qwen2.5-VL-32B: its public weights are AWQ quantised, not comparable with the bf16 of 3B/7B, so we base the scale ladder on the Qwen3-VL family, whose 2B/4B/8B points all have bf16 weights.

3.2.5 Judge

The annotations of the reproduced dataset are long descriptive phrases (median 4 words, maximum 116). Deciding by substring containment systematically errs — when the model answers **Spring** and the annotation is **latewinter to spring**, the model is right but recorded as wrong — so we use bidirectional containment plus content-word coverage tiers, which passes all 22 regression cases. The judge’s three-tier definitions are in Table S6 (online material); in the main experiment the **letter** tier accounts for 1.000 at 3B and 0.984 at 7B, the rest **text**, with **none** below 1%. The free-generation arm has no option letters and relies entirely on alias matching; its criterion defect and correction are in 4.1.1.

3.2.6 A note on baselines

This paper is an analytical work, and the meaning of baseline differs from that in a conventional methods paper. Table 2 lists comparisons of measurement protocols of the same kind, used to show that our conclusions are not the product of one particular measurement convention rather than as "compared methods".

Table 2 Comparison of measurement protocols of the same kind and their role in this paper

Protocol	Source	Role in this paper
Free generation with instruction-style prompts	Reproduced work	Reproduces the known negative sign
<code>ctx_only</code>	Our stance scan	Negative end
<code>neutral</code>	Our stance scan	Near the zero crossing
<code>prior / own_only</code>	Our stance scan	Positive end
<code>len_ctrl</code>	Added in this paper	Rules out the length cause
<code>ctx_hedge</code>	Added in this paper	Covers the deployment intermediate state
ConflictBank template	Existing work	Cross-framework comparison
Explicit and implicit conflict template	Existing work	Cross-framework comparison
Forced-choice format	This paper	A third value of the sign

One point stated firmly: the scale-gradient claim holds only within a single model family, and we make no cross-family comparison. The knowledge-rate difference between the two families belongs to training data and recipe, is unrelated to scale, and would produce meaningless conclusions if mixed in.

Figure 2. The low-decision-rate row on the right is where the denominator collapses, marked with a dashed line. Denominator collapse makes the Δ of that cell uninterpretable, so such cells are excluded from the span computation.

Figure 3. The rescued samples concentrate in a single cell, `ctx_hedge`: answers under that stance often mention both values at once, the alias-only criterion discards the whole answer, and the negation-aware criterion re-judges it as PARAM according to the explicitly negated direction.

4 Experiments and results

4.1 Main result: the sign of the scale effect is decided by the instruction stance

4.1.1 Choice of criterion

Before reporting any scale difference we must fix what counts as "the model followed parametric knowledge". Our dependent variable is a binary choice behaviour requiring a criterion that decides between PARAM and CTX; two candidates exist, and they do not affect the conclusion identically, so the choice needs justification.

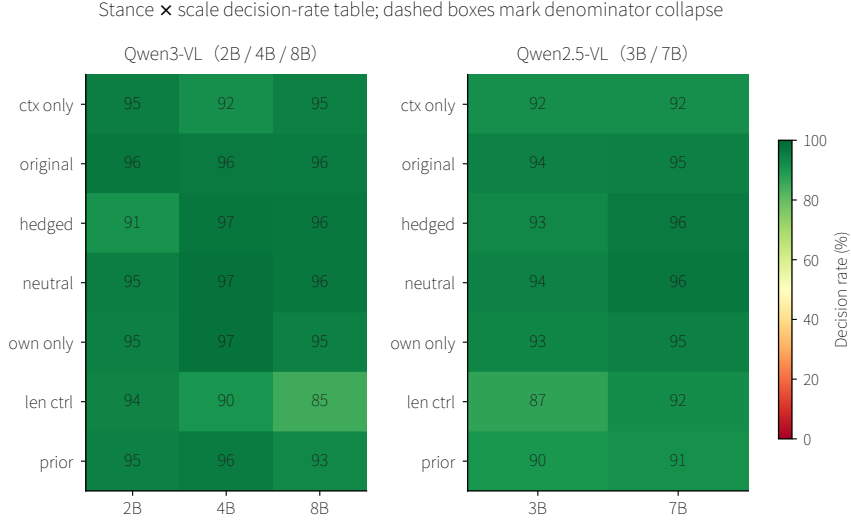


Fig. 2 Full decision-rate matrix over stance $\times$ scale.

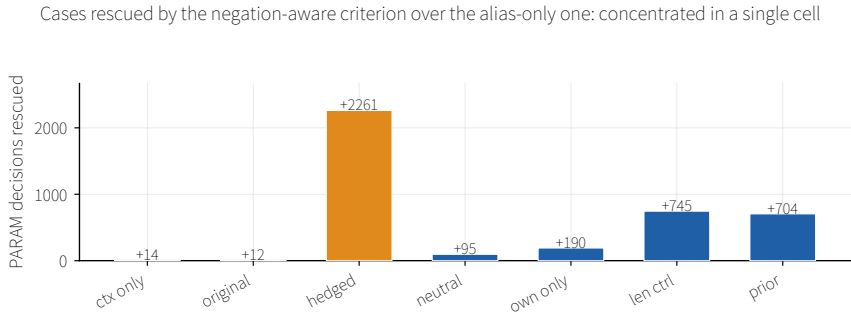


Fig. 3 Cases rescued by the negation-aware criterion over the alias-only one.

The first criterion, **distinctive_ground**, requires all content words of the answer to fall within a single evidence source. Its appeal is strictness: only a model that can actually produce that proper name counts as following the parameter. But the longer the answer, the harder this is to satisfy, so the probability of being judged “undecidable” rises with length. The fragility of string criteria to paraphrase is documented in the evaluation literature and has been shown to be enough to reverse the entire ordering of methods [33, 35]; the length covariation here is a concrete form of the same problem.

The mean answer lengths of the seven stances differ by nearly 20-fold (**orig** 1.5 words, **ctx_hedge** 30.6 words), and the undecidable rate under the original criterion tracks the mean length almost exactly (Table 3):

The Pearson correlation across the seven points is $r = 0.983$, the Spearman rank correlation $\rho = 1.000$. Table 3 is taken from Qwen2.5-VL 3B on the common coverage set. Recomputing on the anchoring set D^* gives $r = 0.986$, $\rho = 1.000$, and the

Table 3 Mean answer length and undecidable rate under the original criterion, by instruction stance

Stance	3B mean length (words)	Undecidable rate under the original criterion
<code>orig</code>	1.5	20.7%
<code>ctx_only</code>	2.2	22.5%
<code>neutral</code>	3.7	30.0%
<code>own_only</code>	6.0	33.9%
<code>prior</code>	17.9	54.3%
<code>len_ctrl</code>	25.8	68.3%
<code>ctx_hedge</code>	30.6	94.4%

conclusion is unchanged; switching to character units also gives $r = 0.983$, $\rho = 1.000$. This covariation is insensitive to the measurement convention. The correlation is between seven stance-level aggregate points, which is the level at which we compare: the difference between stances is precisely a length difference.

This criterion therefore largely measures "how long the answer is" rather than "whose side the model takes": the undecidable rate of `ctx_hedge` under it reaches 94.4%, while most of those are decidable under the alias criterion.

The second criterion matches against two alias tables, `parametric_aliases` and `context_answer_aliases` (`common.answers_match`: bidirectional containment plus content-word coverage). It is length-insensitive, and the alias tables are manually verified, independent of the models evaluated. All main results use it; the comparison with `distinctive_ground` is in 4.4.

Defects of the alias criterion itself, and the second criterion

Bidirectional containment introduces a second, subtler problem. Consider a typical answer:

The context is incorrect. Porrentruy Castle is not located in Peru. It is actually situated in Switzerland.

This answer hits both the context alias Peru and the parametric alias Switzerland and is judged BOTH — "undecidable" — then removed entirely from the PKD denominator. Yet its stance is unambiguous: it rejects the context and gives the parametric ground truth. It should be the strongest evidence for PARAM; the very reason it is removed is that to reject the context it had to name the wrong value.

The defect has a definite direction: the stronger the model, the more it rejects explicitly rather than silently taking a side, so the amount removed rises with scale. On D^* for the `prior` stance the share of "explicit rejections among BOTH" is Qwen3-VL 2B 19.0%, 4B 88.8%, 8B 75.7%, and Qwen2.5-VL 3B 53.1%, 7B 85.8%; the total number of rescues across the seven stances is Qwen3 166 $\rightarrow$ 956 $\rightarrow$ 1042 and Qwen2.5 600 $\rightarrow$ 1257, both rising monotonically with scale. The removal is thus biased against large models and systematically underestimates $\text{PKD}_{s_{\max}}$, pushing down our core positive effect.

We therefore introduce a second criterion: for answers judged BOTH, detect whether they explicitly reject the context and, if so, count them as PARAM. The detection uses a pre-registered table of string patterns in two classes, with the fence-sitting cues (`couldbeeither`, `uncertain`, `botharepossible`, `ambiguous`) never rescued once they appear, so as to prefer missing rescues. The pattern table, its disposition (Table S12), and the two-sided spot-check record (19 rescued answers judged PARAM; 0 of 19 retained falsely rescued) are in the online material.

We report both criteria without selective presentation: Table 4 places all claims of the two side by side, marking the one that is inconsistent.

Table 4 Claims under the alias-only criterion versus the **negation-aware** criterion

Claim	Alias-only criterion	Negation-aware criterion
Number of significant conditions, qwen3 family	6/7	6/7
Number of significant conditions, qwen25 family	6/7	7/7
Span, qwen3 family	45.9 pp	56.5 pp
Span, qwen25 family	37.5 pp	47.5 pp
Sign ordered (all negatives before positives)	Holds for both families	Holds for both families
Δ crosses zero	Holds	Holds
Magnitudes not strictly monotone	Holds	Holds
Cell-wise consistent sign ($n \geq 50$ convention)	Holds	Holds

The correction makes the effect sizes larger, widening the span by 10.6 pp and 10.0 pp in the two families, with the sign structure unaffected. The two criteria disagree in sign in exactly one cell, `ctx_hedge` of the Qwen2.5 family (-4.2 pp, $n = 24$, alias; $+11.6$ pp, $n = 584$, **negation-aware**); it misses the $n \geq 50$ threshold of 4.4 and by rule does not enter the sign comparison, and in all cells that pass, the two criteria agree cell by cell. The denominator gap (24 versus 584) is itself an instance of our argument: the alias criterion removes answers that must name the wrong value in order to reject the context, and `ctx_hedge` most strongly induces rejection, so it suffers most (4.4.2, 4.6.2). The criterion is a string rule, not a semantic decision; complex or ironic phrasing may be missed, but a miss only falls back to the alias-only criterion and cannot manufacture false positives, because a rescue requires an explicit pattern match. The spot-check records for both blind directions are in the online material.

4.1.2 Stance ladder

The sign of the scale effect depends on where the instruction places parametric knowledge. We set seven stances and pre-register their ordering from weak to strong authorisation — pre-registered meaning the ordering is fixed before the data are seen,

not derived from a post hoc fit. Conclusions under a single prompt often cannot represent behaviour over the prompt space [35, 36], and authorisation strength is a prompt dimension not previously scanned on its own. The wording and authorisation strength are in Table S13 (online material; verbatim prompts in Table 1). The position of `len_ctrl` deserves separate comment: it separates "authorise parametric knowledge" from "require a long answer", and is our main design for countering the length confound. Were the scale effect driven purely by answer length, `len_ctrl` should give results close to `prior`; empirically it does not (4.1.4).

4.1.3 Cross-scale results

For each stance ι and scale s , the PKD rate is defined as

$$\text{PKD}_s(\iota) = \frac{|\{i \in D^* : \text{grade}(y_s(i, \iota)) = \text{PARAM}\}|}{|\{i \in D^* : \text{grade}(y_s(i, \iota)) \in \{\text{PARAM}, \text{CTX}\}\}|}$$

The denominator counts only decidable samples; BOTH and NONE are removed. The scale effect is defined as the difference between the largest and smallest scale within a family:

$$\Delta_s(\iota) = \text{PKD}_{s_{\max}}(\iota) - \text{PKD}_{s_{\min}}(\iota) \quad (1)$$

Measured results (5 scale points, two model families)

The number of scale points expanded from 2 to 5: Qwen3-VL forms the 2B→4B→8B doubling ladder ($|D^*| = 649$) and Qwen2.5-VL a second family, 3B→7B. Table 5 gives the endpoint comparison for the two families under the **negation**-aware criterion.

Table 5 PKD endpoint comparison by stance for the two model families (**negation**-aware criterion)

Stance	Qwen3 2B	Qwen3 8B	Δ_{q3} (pp)	p
<code>ctx_only</code>	0.324	0.050	−27.3	2.9×10^{-38}
<code>orig</code>	0.382	0.145	−23.7	2.6×10^{-29}
<code>ctx_hedge</code>	0.687	0.979	+29.2	4.5×10^{-49}
<code>neutral</code>	0.549	0.564	+1.5	0.491
<code>own_only</code>	0.621	0.742	+12.1	4.9×10^{-10}
<code>len_ctrl</code>	0.548	0.762	+21.4	1.9×10^{-26}
<code>prior</code>	0.631	0.911	+28.0	8.3×10^{-45}

The two families independently give an isomorphic conclusion: the two weakest-authorisation stances have $\Delta < 0$, the two strongest have $\Delta > 0$, and Δ crosses zero — two different families, different parameter ranges, run independently, with the same signs.

Table 6 PKD endpoint comparison by stance, Qwen2.5-VL (negation-aware criterion)

Stance	Qwen2.5 3B	Qwen2.5 7B	Δ_{q25} (pp)	p
<code>ctx_only</code>	0.244	0.160	-8.4	2.5×10^{-6}
<code>orig</code>	0.347	0.209	-13.8	4.5×10^{-13}
<code>ctx_hedge</code>	0.846	0.962	+11.6	1.1×10^{-18}
<code>neutral</code>	0.457	0.615	+15.9	8.7×10^{-15}
<code>own_only</code>	0.449	0.666	+21.7	4.4×10^{-25}
<code>len_ctrl</code>	0.572	0.694	+12.3	2.5×10^{-10}
<code>prior</code>	0.552	0.890	+33.7	3.5×10^{-50}

The effect does not grow smoothly with scale but saturates early

The decomposition into adjacent pairs yields an important qualification. For the Qwen3 family, on the three strongest-authorisation stances the 2B→4B segment already achieves almost the entire effect and the 4B→8B segment is essentially flat (`ctx_hedge` +2.5, `len_ctrl` -0.6, `prior` -1.0, the latter two not significant): the scale effect saturates early within our range and does not grow smoothly with scale. The shape of the curve depends on the measure — the same quantity can appear as a continuous climb or a discontinuous jump depending on difficulty and scoring convention [26, 27] — and we measure the sign of a binary decision, so the saturation point may precede that of the capability itself. The correct statement is: the sign is decided by the stance, the magnitude saturates early. The two families differ in shape: the negative side of the Qwen3 family does not saturate (`ctx_only` -12.9 then -14.7), whereas the Qwen2.5 family is near saturation at 7B. The adjacent-pair values are in Table S2 (online material).

“Sign ordering” and “monotone magnitudes” are two different things and must not be conflated

The two claims have different evidential strength and must be split. Weak claim (supported): the sign of Δ is ordered with authorisation strength, all negative signs preceding all positive signs with no interleaving. Strong claim (not supported): the magnitude rises monotonically with authorisation strength. Qwen2.5-VL 3B→7B has two adjacent inversions (`ctx_only` -8.4 $\geq$ `orig` -13.8, `own_only` +21.7 $\geq$ `len_ctrl` +12.3), $\rho = 0.857$ ($p = 0.012$); Qwen3-VL 2B→8B has one (`ctx_hedge` +29.2 $\geq$ `neutral` +1.5), $\rho = 0.643$ ($p = 0.070$). The trend is significant but far from monotone, so we claim only the weak claim, which holds under both criteria (paired convention, spans 45.9/37.5 pp).

Sensitivity check excluding ctx_hedge

`ctx_hedge` is the only one of the seven cells whose magnitude is highly unstable across criteria (the two criteria differ by 60.1 pp here), so it is worth testing what remains when it is removed entirely. The weak claim not only holds but becomes stronger. The rank correlations and spans before and after exclusion are in Table S3 (online

material). The sign sequences of the two families are identical after exclusion — still the first two negative and the last four positive, still crossing zero — and the span is nearly unchanged (Qwen3 drops only from 56.5 to 55.3 pp); the Qwen3 family becomes strictly monotone ($\rho = 1.000$), so its low rank correlation with seven cells is caused entirely by **ctx_hedge**. Two implications follow: the least credible cell is not load-bearing — remove it and the conclusion is strengthened, not weakened — and **ctx_hedge**’s absolute magnitude overlaps heavily with the **negation** criterion’s pattern table, dragging the Qwen3 family’s rank correlation from 1.000 down to 0.643. We therefore keep **ctx_hedge** in the table for the complete seven-level scan but rely on it to support no claim. The Qwen2.5 family’s ρ drops from 0.857 to 0.771 after exclusion, its monotonicity still limited by the adjacent inversion **own_only** (+21.7) $\geq$ **len_ctrl** (+12.3), which excluding **ctx_hedge** cannot fix; the two families’ rank correlations go one up and one down, so our weak claim remains limited to sign ordering and we do not claim monotone magnitudes.

len_ctrl: authorisation decides the sign, length decides the magnitude

Among the three inversions above, **own_only** $\geq$ **len_ctrl** is the most informative, being a built-in control: both stances’ authorisation wording requires parametric knowledge to take precedence, and only **len_ctrl** additionally constrains answer length. On Qwen2.5-VL 3B $\rightarrow$ 7B it lowers the length by 35 characters while pushing Δ from +21.7 down to +12.3. The determinants of sign and of magnitude are in Table S14 (online material). This is complementary to 4.1.4’s refutation of H_{len} , not contradictory: length cannot decide the sign, but it can decide the magnitude — hence the paper limits its title-level assertion to the sign and attributes the magnitude explicitly to length, a known confound.

4.1.4 Ruling out the length confound

A natural alternative explanation is that large models give longer answers, and longer answers have more opportunities to hit a parametric entity word, so that ”following parametric knowledge more” is really length at work. Denote this H_{len} . It has a directly decidable corollary: if it holds, the sign of Δ_s should follow the sign of the large-minus-small answer-length difference,

$$\text{sgn} \left(\overline{|y_{s_{\max}}|} - \overline{|y_{s_{\min}}|} \right) = \text{sgn} \Delta_s(\iota), \quad \forall \iota$$

Any cell with ”length grows while $\Delta < 0$ ” or ”length shrinks while $\Delta > 0$ ” is a counterexample. Table 7 gives the decision for each of the fourteen cells of the two families, with Δ from the **negation**-aware criterion (cognate with 4.1.3): the length difference, Δ , the decision rate, and the predicted versus measured sign of H_{len} ; cells where the two disagree are counterexamples.

Six of the fourteen cells have opposite signs, refuting H_{len} . The sharpest are **ctx_only** and **orig** of the Qwen2.5 family: the large model gives longer answers yet follows parametric knowledge less. And **len_ctrl**, designed specifically to separate authorisation from length, does lower length by about 23 characters while Δ remains positive. Under the alias-only criterion there are more opposite-sign cells (eight of fourteen), because it flips the length differences of the **ctx_hedge**, **own_only**, and **len_ctrl**

Table 7 Length-confound decision table for the two families (**negation-aware** criterion)

Family	Stance	Length diff (chars)	Δ (pp)	Decision rate	H_{len} pred.	emp.
Qwen3	ctx_only	-7.7	-27.3	91.8%	negative	negative
Qwen3	orig	-2.3	-23.7	93.7%	negative	negative
Qwen3	ctx_hedge	+68.9	+29.2	88.1%	positive	positive
	e					
Qwen3	neutral	-14.8	+1.5	92.9%	negative	positive
Qwen3	own_only	-4.7	+12.1	90.8%	negative	positive
Qwen3	len_ctrl	+83.1	+21.4	81.5%	positive	positive
Qwen3	prior	+48.2	+28.0	88.6%	positive	positive
Qwen2.5	ctx_only	+15.0	-8.4	86.4%	positive	negative
Qwen2.5	orig	+2.3	-13.8	91.4%	positive	negative
Qwen2.5	ctx_hedge	-35.5	+11.6	90.0%	negative	positive
	e					
Qwen2.5	neutral	+37.6	+15.9	92.1%	positive	positive
Qwen2.5	own_only	+32.4	+21.7	90.0%	positive	positive
Qwen2.5	len_ctrl	-22.6	+12.3	81.7%	negative	positive
Qwen2.5	prior	+42.7	+33.7	83.7%	positive	positive

cells to negative — all three with denominator collapse (decision rate 13%–40%), which records "the length of the removed part" as a negative difference and is not comparable, for the same reason as in 4.1.3. H_{len} is refuted under both criteria, in the same direction. Stance does change length (Kruskal-Wallis $p < 10^{-160}$ on every scale point, $H = 1764.5/1915.3/1495.4/2344.7/2461.0$, $df = 6$), so length is not a constant and H_{len} is not untestable; it is simply not sufficient to decide the sign.

A corollary: the decision rate must be reported alongside Δ

The defects of our criterion first surfaced in the decision rate. **ctx_hedge** at 3B has a mean length of 188 characters, 20 times that of the adjacent stance **orig**, because that prompt makes the model give the parametric and the context answer at once; under the alias-only criterion such answers fall into BOTH and are removed, so its decision rate is as low as 6.0%–66.3% while other stances are mostly above 80%. The decision rate must therefore be given alongside Δ : reporting only Δ and p places a weakly evidenced condition beside a strongly evidenced one, and every table here that reports Δ includes a decision-rate column. The **negation-aware** criterion lifts this column across the board (**ctx_hedge** from 6.0% to above 90%, the whole table stabilising at 85%–97%), direct evidence that it repairs the denominator collapse — the low decision rate is not "the model cannot answer" but "the model rejected and was removed" — without removing the reporting standard.

4.1.5 Shrinkage of the anchoring protocol

The 927-sample main set was itself filtered by the intersection of the knowledge of Qwen2.5-VL 3B and 7B, so for these two $K_s(i) = 1$ holds almost by construction; the newly added Qwen3-VL family needs independent testing. The signal is unambiguous:

Qwen3-VL 2B produces the parametric ground truth under NO-CTX only 0.785 of the time, against 0.992/0.997 for Qwen2.5-VL 3B/7B. This must be handled, or the comparison is contaminated by a trivial effect: if the small model does not know a fact, "not following parametric knowledge" is not a choice but having nothing to follow, and leaving it in the denominator would systematically favour the small model. Note this is opposite in direction to the criterion defect of 4.1.1 — that pushes down large models, this pushes down small ones, and the presence of both does not mean they cancel.

The anchoring protocol takes the samples that all scale points answer correctly:

$$D^* = \bigcap_s \{i : K_s(i) = 1\} \quad (2)$$

where $K_s(i)$ is measured on the NO-CTX arm alone, independent of the conflict condition. D^* shrinks monotonically as scale points are added: 720 for the earliest three scale points (Qwen3-VL 2B, Qwen2.5-VL 3B/7B), 672 after adding Qwen3-VL 4B, and 649 when all five scale points are present, 70.0% of the 927 common-coverage samples. The amount removed for each model relative to these 927 is in Table S4 (online material).

The amount removed falls monotonically with scale, consistent with the knowledge rate rising with scale (relative to the common coverage of 927, not to each model's own knowledge set). D^* shrinks from 927 to 649, a loss of 30%, but 649 still supports every test here (the per-cell decidable samples are above 550), so the scale conclusion remains confirmatory. The main cost falls on Qwen3-VL 2B, 21.7% of whose samples are removed: its PKD is measured on the subset it genuinely knows, so the remaining difference can be attributed to scale and stance rather than to how much is known.

Figure 4. The upper panel plots PKD against scale, one line per instruction stance; the slope of each line is the direction of the scale effect under that stance. The lower panel gives Δ per stance with its significance, and annotates the number of significant cells and the span. Read together they show the sign flipping with authorisation strength.

Figure 5. The upper panel shows sign crossing: the same stance falls on opposite sides of the zero line under the two criteria, so the criterion changes the sign of the conclusion rather than only its magnitude. The lower panel is the length control: if the effect were driven by strong-authorisation stances answering at greater length, the two axes would show a systematic slope; the measured correlation is close to zero in both families.

Figure 6. (a) Where the same batch of answers goes under the two criteria. BOTH under the alias-only criterion means undecidable, and such answers are removed from the PKD denominator altogether. (b) The share of BOTH that is explicit rejection under the **prior** stance, all five scale points plotted. In Qwen3 the share peaks at 4B and then falls (4B to 8B: -13.1 pp), the only non-monotone point recorded in 4.1.1; the absolute number of rescued items rises monotonically with scale within both families.

Figure 7. Ordering the stances by authorisation strength, PKD rises overall, but **ctx_hedge** lands at the top, a deviation that must be reported.

PKD rate vs. model scale under each instruction stance (negation-aware criterion, $|D^*| = 649$)

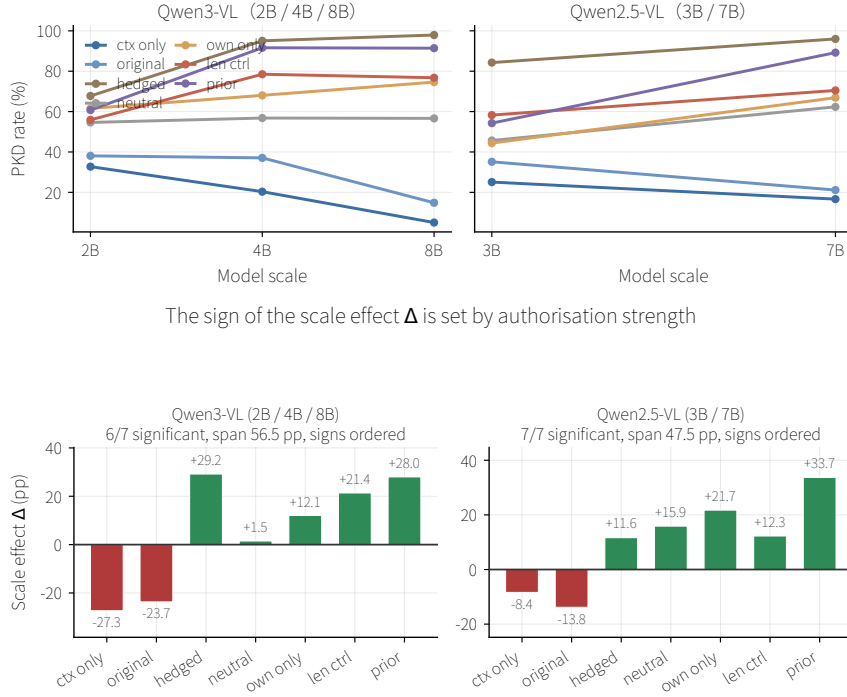


Fig. 4 Interaction of instruction stance with scale, and the sign of the scale effect Δ .

4.2 Framework replication and self-refutation

All the conclusions of 4.1 rest on a question format of our own design. One obvious objection is: could the conclusions simply be an artifact of our question format? Prior work has reported this phenomenon using its own different templates; if we reproduced them verbatim, would we obtain the opposite sign, thereby showing that our gradient does not generalize?

This section tests that question directly. The conclusion is that our core finding is robust to the choice of framework, but the replication process exposed one failure that must be reported: the forced-choice template of one framework is completely dominated by position preference, and the numbers under that condition cannot be used for any content-level inference.

4.2.1 The three replicated external frameworks

The replication targets are the templates of two published works, whose wording and option construction we copied verbatim. The sources and construction of the three external framework conditions are given in Table S7 (online material).

In addition, that work’s default condition `cb_default` serves as a control. The four conditions share the same set of 927 samples, the same pair of models (Qwen2.5-VL

Sign crossing: under the alias-only criterion the length-control stance escapes the negative region in Qwen3

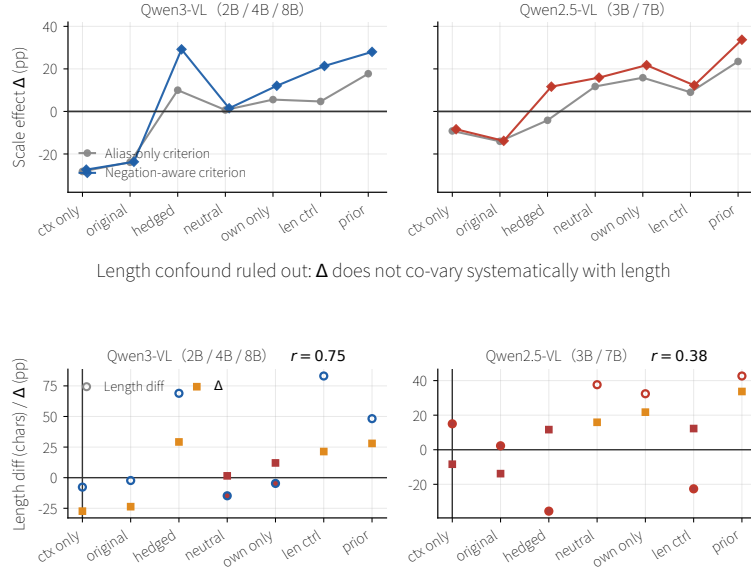


Fig. 5 Sign crossing and the exclusion of the length confound.

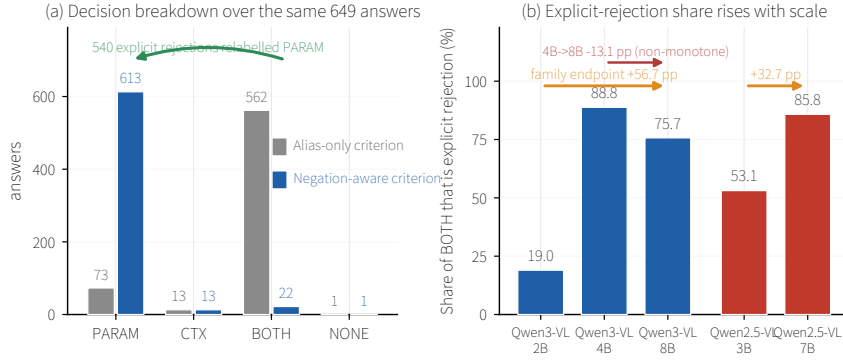


Fig. 6 The rescue mechanism of the negation-aware criterion: the more explicit the rejection, the more likely the alias-only criterion is to discard it.

3B/7B), and the same decoding parameters. The motivation for this section comes from a repeatedly documented observation: option order and option format both shift multiple-choice results substantially [32, 33], so "framework invariance" itself needs to be tested rather than assumed.

Stance gradient: PKD rises with authorisation strength; the hedged stance escapes the ladder

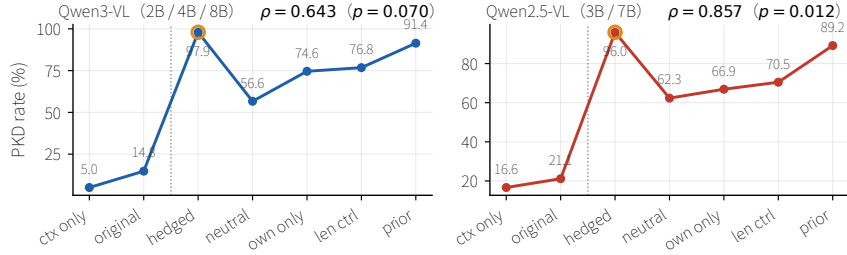


Fig. 7 Stance gradient: PKD rises with authorisation strength, and `ctx.hedge` escapes the ladder.

4.2.2 Main result: the sign does not flip with the framework

Under the replication run with a fixed option order (the parameter’s true value is always A), the size effect $\Delta P(A)$ is negative for all four conditions:

Table 8 Size effects and McNemar tests for the four replication conditions

Condition	3B	7B	$\Delta P(A)$	McNemar p
cb_default	0.981	0.941	-0.040	3.0×10^{-9}
cb_conflict	0.494	0.475	-0.019	0.23
xie_implicit	0.479	0.476	-0.003	0.89
xie_explicit	0.475	0.436	-0.039	0.018

This agrees in sign with the two weakly authorized stances `ctx.only` and `orig` of our 4.1. In other words: under formatted forced-choice questions, the sign of the size effect is stably negative and does not change with the framework. The disagreement between the two published works cannot be explained by their prompt frameworks alone, which is consistent with the stance conclusion of our 4.1 — what determines the sign is the locus of authorization over parameter knowledge, not the template wording.

The magnitude of the effect should be noted: under forced choice it is at most only 4 pp, whereas under free generation the stance sweep spans 47.5 to 56.5 pp (negation-aware criterion, Qwen2.5 family and Qwen3 family). The format itself compresses the effect, and this is itself the “third value of the sign” to be reported in 4.3.

4.2.3 One replication failure: position preference can completely override content

The reliability of the table above depends on one premise: the option letter carries no information. We test it by swapping option positions, rerunning the same samples with the A/B contents exchanged. (Sequence position also changes how much the

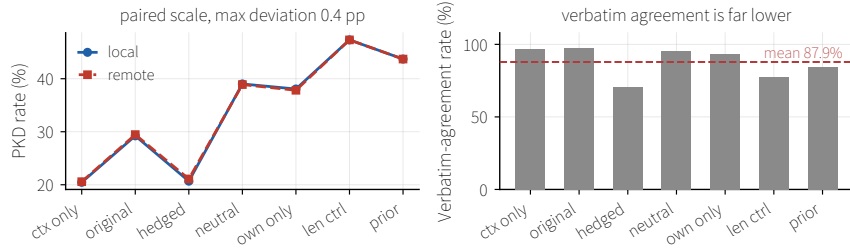


Fig. 8 Cross-environment replication.

model exploits the same content [44]; our swap conflates order and position effects and cannot fully separate them, a limitation we list.) Position preference is independently reported in multiple-choice evaluation and model judging [32, 33, 37]; our contribution is its magnitude under the conflict condition — 0.961 is close to complete domination. The premise fails completely for `cb_default`, whose position preference is close to 1 (Table S8, online material): its output is almost entirely determined by which letter is the answer, independent of content, appearing to nearly always choose correctly under one order (0.981) and incorrectly under the other (0.019). Neither number measures “model judgment”, and `cb_default` cannot be used for any content-level inference. The position preferences of `cb_conflict` and both Xie frameworks are all below 0.11, so content dominates there and the sign conclusions of 4.2.2 hold; the `cb_default` row, though of the same sign, must not be cited as evidence.

4.2.4 The self-refutation role of this section

We keep this failure in the main text rather than an appendix because it shows our robustness checks are effective rather than formal: without the order swap, `cb_default`’s 0.981 would be read as “the model makes almost no errors in this format” and, added to our negative-sign conclusion, would reinforce a false impression. It also yields a transferable warning: the accuracy of a forced-choice format is uninterpretable before order is controlled, yet such formats are widely used in the knowledge-conflict literature and position preference is invisible without a control. We therefore report order-control results for every forced-choice condition, using their numbers only where position preference is negligible.

Figure 8. The left panel gives PKD under the paired convention, where the maximum deviation is limited; the right panel gives verbatim agreement, where string-level replication is far worse. Side by side they show that a replicable conclusion and replicable output are two different things.

4.3 Statistical tests and power

All of our comparisons are paired binary comparisons: the same samples receive different conditions within the same model and run, and what is compared is the direction of the flip. This determines the choice of test and the power calculation.

4.3.1 Primary test: McNemar exact test

For each condition pair we construct a 2×2 contingency table; only the discordant cells (the two directions of flip) enter the test. We use the exact binomial version of McNemar rather than the chi-square approximation, because the flip counts are small in some cells (`cb.default` only 43) and the approximation is unreliable there: for discordant counts b and c , $p = 2 \sum_{i=0}^{\min(b,c)} \binom{b+c}{i} 2^{-(b+c)}$, truncated at 1, with $p = 1$ when the flip count is zero. We give the effect size alongside the sample size rather than significance alone [40]. A paired test is chosen over a two-group independent-proportion test for power: the same samples’ answers under the two conditions are highly correlated, so the paired design removes the large variance component of sample difficulty. For example, `prior` on Qwen2.5-VL 3B $\rightarrow$ 7B has flips of 15 versus 234, giving $p = 1.0 \times 10^{-51}$ from only 249 discordant samples.

4.3.2 Twofold testing of monotonicity

4.1 needs to test the ordered alternative “ Δ rises with authorization strength”, which cannot use a chi-square-type test, so we use two complementary quantities: the Spearman rank correlation ρ , measuring the strength of the trend, and a permutation test with 2×10^5 reshufflings of the authorization-order labels, giving the null distribution of ρ and the one-tailed p . Permutation rather than a table lookup is used because with only 7 stances the asymptotic distribution is unreliable; the test makes no distributional assumption, and 2×10^5 reshufflings suffice to estimate p stably to three significant figures. Both are reported because they answer different questions — ρ “how strong is the trend”, p “how likely is a trend this strong by chance” — and we have cases where ρ is significantly positive yet the ordering still contains inversions (4.1.3), where reporting only “significant” would cause overinterpretation.

4.3.3 Between-group comparison: Kruskal-Wallis

Testing whether “stance significantly changes answer length” compares the length distribution of the seven stance groups. Such data violate normality — the answers are text, the distribution is right-skewed, and the median often falls in single digits while the P90 exceeds 300 — so we use the Kruskal-Wallis rank-sum test, which requires only similar distribution shapes. With R_j the rank sum of group j ,

$$H = \frac{12}{N(N+1)} \sum_{j=1}^k \frac{R_j^2}{n_j} - 3(N+1), \quad H_{\text{corr}} = H / \left(1 - \frac{\sum (t^3 - t)}{N^3 - N} \right)$$

the second being the tie correction, which cannot be omitted on integer length data. The tail probability is given by the chi-square distribution; the implementation and its validation are described in the reproduction material.

4.3.4 Power accounting

The anchor set D^* has 649 items and all primary comparisons are paired on the same samples, so the detection limit must be given for a paired design. With π_d the observed proportion of sample pairs on which the two models disagree, the minimum detectable effect is

$$\text{MDE} \approx (z_{1-\alpha/2} + z_{1-\beta}) \sqrt{\frac{\pi_d}{n}} \quad (3)$$

Because π_d is observed rather than designed, this accounting is post hoc: it answers "given the disagreement rate actually observed for each stance, how large a Δ can be detected", not "how many samples should have been recruited". Post hoc detection limits carry a known risk of being over-optimistic [28], since the disagreement rate is itself affected by the condition, so these numbers serve only as a sieve for "which cells are adequately powered", not as a power claim.

Taking $\alpha = 0.05$ and power $1 - \beta = 0.80$, and substituting the observed π_d for each stance (range 0.123 to 0.351), we obtain the per-stance detection limits: 4.1 to 7.1 pp for the Qwen2.5-VL family, median 5.5 pp; and 5.4 to 6.4 pp for the Qwen3-VL family, median 6.3 pp. All values were computed by `src/analyze_mde.py` and saved to `results/mde.json`.

Read alongside the per-cell results of 4.1: the smallest significant Δ across the two families is -8.4 pp for `ctx_only` in the Qwen2.5-VL family, exceeding that stance's 5.0 pp detection limit, and the other significant cells are larger, up to 56.5 pp (between $+29.2$ for Qwen3-family `ctx_hedge` and -27.3 for `ctx_only`). "Significant" on these cells is therefore not an artifact of sample size.

Only one cell falls below its detection limit: $\Delta = +1.5$ pp with $p = 0.491$ for the Qwen3-family `neutral` stance, below that stance's 5.4 pp detection limit. This non-significance does not mean the effect is zero; it is below the detection limit. Throughout the main text we use "not detectable at our sample size" for non-significant cells rather than "no effect," and we neither use a sub-limit effect as evidence nor interpret it as "no difference in fact."

4.3.5 Two common tests we do not use

All primary comparisons are paired on the same samples, so we do not use the independent-sample proportion test (it would underestimate power and overestimate p). The Stuart-Maxwell test applies to paired comparisons among A/B/C; our abstention-pivot claim was falsified and retracted, so the test lost its use and we report no abstention conclusion based on three-category pairing.

4.4 Criterion comparison

Our main results depend on one set of alias-matching criteria. This section reports all alternative criteria side by side, so that readers can judge how much of the conclusion comes from the choice of criterion. The three agree in direction but differ greatly in magnitude. Criterion dependence cannot be adjudicated by how good the results are: the reasons must be written down before the results are seen [35, 36], otherwise any choice can be defended after the fact.

4.4.1 The three criteria

The defect of **distinctive_ground** (its undecidable rate covaries with answer length, $r = 0.983$) was described in 4.1.1; we retain it only as a control. Table 9 gives the decision rule and length sensitivity of the three criteria.

Table 9 Decision rules and length sensitivity of the three criteria

Criterion	Decision rule	Length-sensitive
distinctive_ground	All content words of the answer must fall within a single evidence source	Highly sensitive
Alias matching	Bidirectional containment with the parameter alias table or the context alias table	Not sensitive
Negation-aware alias	Alias matching + those explicitly rejecting the context count as following the parameter	Not sensitive

4.4.2 Comparison results for the three criteria

We compare the Δ given by the three criteria on the same family (Qwen2.5-VL 3B $\rightarrow$ 7B). The numbers in parentheses are the paired decidable sample count for that cell — both sides must be decidable.

On the six cells with sufficient paired decidable samples the three criteria agree completely in sign, with no cell reversed: the sign conclusion does not depend on the criterion. The only sign-reversed cell is **ctx_hedge**, precisely where the denominator collapses most severely. The denominator collapses under the two length-sensitive criteria and is restored under the **negation**-aware one (Table 11):

The three rows say the same thing: the denominator collapses from 584 to 1, not because the criteria "disagree" but because the same answer is judged undecidable; after the collapse Δ can only take degenerate values of ± 100 (the -100.0 above is determined by a single sample), yet in the table it looks like "the cell with the strongest effect". We therefore stipulate that cells with $n < 50$ are not counted in the span and take part in no sign comparison (as in 4.5.2: whenever a proportion is reported, its denominator must accompany it). Without this the table would create the illusion that "the three criteria disagree on **ctx_hedge**", whereas two of them have no usable denominator there at all; "six cells consistent" is the robustness claim we actually use. In magnitude the **negation**-aware criterion gives the largest span (47.5 pp) and alias

Table 10 Size effects given by the three criteria on the same model family

Stance	distinctive_ground	Alias matching	Negation-aware
ctx_only	-7.3 ($n = 451$)	-9.2 ($n = 556$)	-8.4 ($n = 561$)
orig	-11.7 ($n = 496$)	-14.0 ($n = 591$)	-13.8 ($n = 593$)
ctx_hedge	-100.0 ($n = 1$, insufficient denominator)	-4.2 ($n = 24$, insufficient denominator)	+11.6 ($n = 584$)
neutral	+8.7 ($n = 345$)	+11.7 ($n = 520$)	+15.9 ($n = 598$)
own_only	+15.0 ($n = 327$)	+15.9 ($n = 492$)	+21.7 ($n = 584$)
len_ctrl	+16.4 ($n = 122$)	+9.0 ($n = 366$)	+12.3 ($n = 530$)
prior	+28.0 ($n = 93$)	+23.5 ($n = 230$)	+33.7 ($n = 543$)
Span (cells with $n \geq 50$ only)	39.7	37.5	47.5

Table 11 Paired decidable sample counts and discarded amounts for the **ctx_hedge** stance under each criterion

Criterion	ctx_hedge paired decidable n	Discarded BOTH count (both sides combined)
distinctive_ground	1	3
Alias matching	24	1108
Negation-aware	584	62

matching the smallest (37.5 pp), consistent with the criteria’s length sensitivity; the larger span of **distinctive_ground** is not ”a larger effect” but ”the hard-to-measure end was deleted” — it leaves only 93 decidable samples on **prior** and 1 on **ctx_hedge**.

4.4.3 The **len_ctrl** cell: where the three criteria diverge most

The three criteria differ most on **len_ctrl** (+16.4, +9.0, +12.3; the n of **distinctive_ground** has dropped to 122, so its magnitude has limited reference value), because it was designed precisely for length control and its answer lengths are of the same order as its authorization stance. Under that stance the model often mentions the parameter value and the context value at the same time, which the pure alias criterion judges BOTH and discards (the judged rate falls to 42.2%–67.7%); the **negation-aware** criterion rescues the explicitly rejecting portion, raising it to

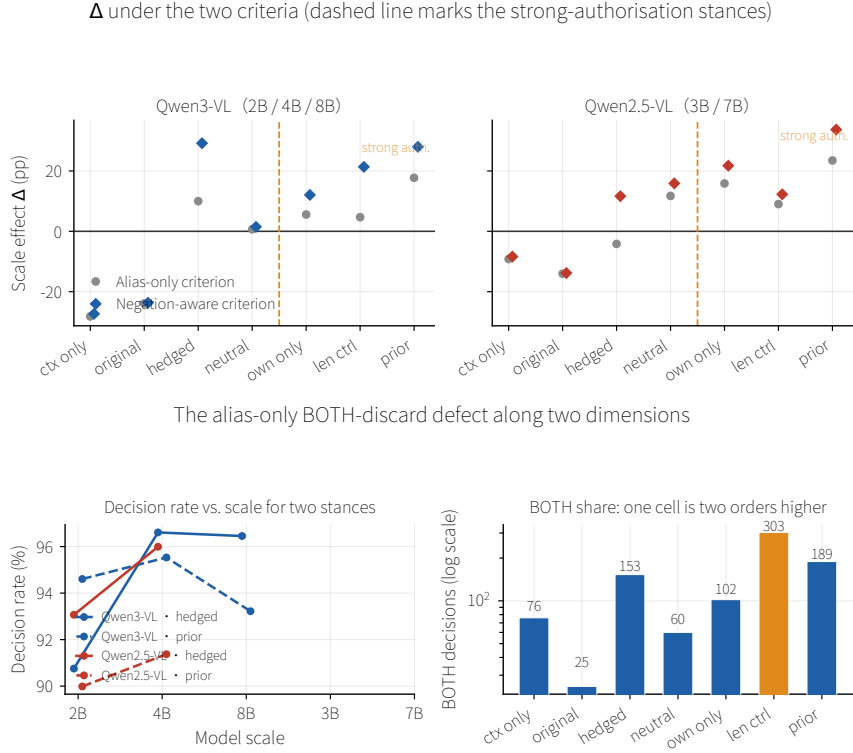


Fig. 9 Δ under the two criteria, and the BOTH-discard defect.

82.8%~89.4%. All three agree in sign, positive; the divergence is only in magnitude, so we claim only the sign of that cell (written into 4.1.3).

4.4.4 Criterion choice cannot be decided by how good the results are

Changing the criterion makes the results stronger, which is itself a danger signal, so our reasons must be structural defects diagnosed in advance and independent of the result’s direction. Two reasons (see 4.1.1): the construction itself fixes the outcome — bidirectional containment forces “naming the wrong value in order to reject it” into BOTH — and the correction rules (the pattern table of `regrade_negation.py`) were fixed before the results were seen, split into five linguistic categories, with no tuning to results. Both criterion sets’ numbers are reported in full in 4.1.1, with no selective presentation. The criterion is a string rule, not a semantic judgment; complex phrasings may be missed, which only falls back to the pure alias criterion and cannot create false positives.

Figure 9. The upper panel contrasts Δ under the two criteria (grey for the alias-only criterion, colour for the negation-aware one); after the switch the dose-response curve becomes too orderly and the span changes, and that difference is itself evidence of a criterion defect. The lower panel gives two dimensions of the BOTH defect: left,

the decision rate; right, the BOTH share, which is two orders of magnitude higher in the `ctx_hedge` cell.

4.5 Stratified test

One explanation to rule out is that the scale effect is carried by a single editing type: if the effect holds within only one slot, it may reflect a property of that slot rather than a general interaction between scale and stance. This section recomputes the scale effect within each slot. The risk of stratification is that a subset effect and the full-sample effect can have the same direction but different significance, and reporting only one is misleading [28, 40].

4.5.1 Slot distribution and results inside the two large slots

The slots are the semantic class of the word replaced by the counterfactual edit; within D^* only `place` and `other` have sufficient sample size, the other three comprising 28 items in total. We compute Δ within each slot under the paired protocol (`negation-aware` criterion, as in 4.1). The Qwen2.5 family’s `place` slot is fully consistent with the overall result, all seven stances significant, span 54.1 pp — the most complete single-slot replication here — while its `other` slot is not, the `ctx_only` cell being +1.1 ($p = 0.832$) and by 4.3.4 given no directional interpretation. The Qwen3 family is the opposite: all seven `other` cells are significant and consistent (span 62.1 pp), whereas the two middle `place` cells are not ($p = 0.440, 0.111$). The only cross-family inconsistency is `other/ctx_only` (Qwen2.5 +1.1, not significant; Qwen3 -7.6, $p = 0.014$); that cell uses the same batch with identical composition in both families, so slot composition cannot explain it, and we record it faithfully as a limitation. The decidable samples for `ctx_hedge` within the two slots are ample (374 and 193) against 12 and 8 under the pure alias criterion — the latter from denominator collapse, not from the stance’s true decidable volume.

4.5.2 Small slots: explicitly no conclusions drawn

For the three classes `diet`, `time`, and `quantity` the decidable samples within the anchoring set drop to the teens or single digits; they are listed only to show that they are unusable (Table S9, online material). The Δ values there can reach +78.6 pp numerically (`prior` of `diet`), but rest on one or two flip pairs, with confidence intervals covering the entire $[0, 1]$. Most cells in the `time` slot are 0.0 — not “no effect”, but neither scale flipping, so the denominator supplies no information. All three small slots are far below the $n < 50$ threshold of 4.4.2, so this section only reports numbers and draws no conclusions. Reporting proportions without denominators gives the mistaken impression that “all slots point the same way and some have stronger effects”; we therefore require the number of flipped cells in all stratified tables.

4.5.3 Relation to differences in slot composition

The slot-composition difference between the two generations (3.1.5) does not affect the scale conclusions: the comparisons are paired, so slot composition is identical on both

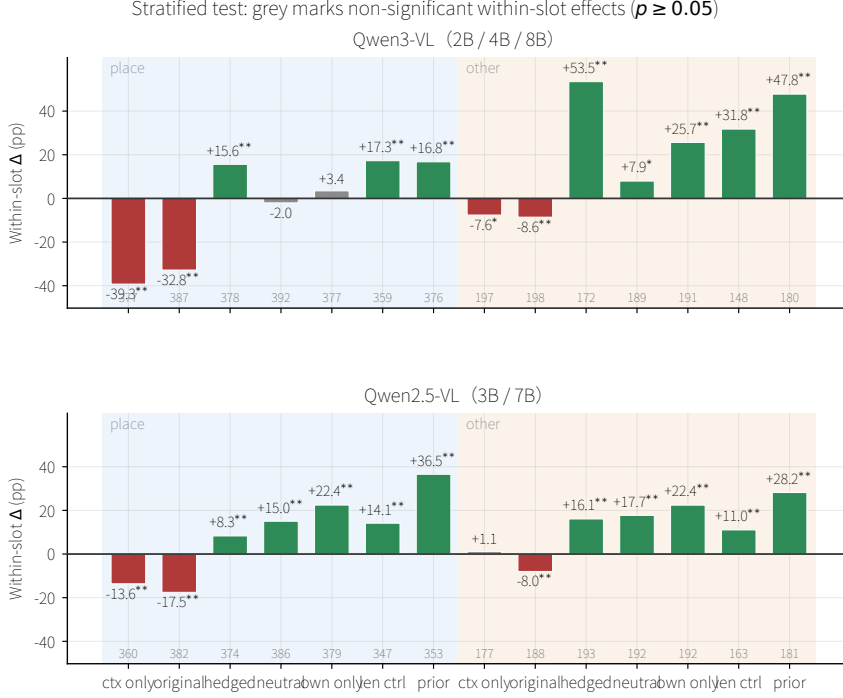


Fig. 10 Stratified test: grey marks non-significant within-slot effects ($p \geq 0.05$).

sides and cancels in the paired difference. It does affect cross-generation comparisons, so we report no cross-generation numerical comparison.

Figure 10. After stratification the sample size inside each slot falls, and only the two large slots have adequate power. No conclusions are drawn for the small slots.

4.6 Limitations

This section lists the boundaries of this paper. The items listed first directly weaken the strength of the claims; the reason they are written in the main text rather than left to be discovered later is that several of them are themselves part of the methodological conclusions of this paper.

4.6.1 Limitations of samples and scale

Only two model families: the scale ladder is built on Qwen3-VL 2B/4B/8B and Qwen2.5-VL 3B/7B, both from the Qwen series, so we cannot rule out that the conclusions are tied to that series' training recipe and we make no cross-family generalization. Evaluation under multiple prompts and conditions has been recommended as the default protocol [35, 36], and our two-family, two-criterion design is the minimal implementation of that recommendation here. The third scale point, 32B, was empirically falsified: its public weights are AWQ quantized, incomparable with the bf16 of the other scales, which would reintroduce the confound of 3.2.4 and break

the anchoring protocol’s precondition. The effect saturates early (4.1.3), so the scale conclusions cover only 2B to 8B and cannot be extrapolated to “the effect keeps growing with scale”. Single data source: all experiments rest on one counterfactual-editing dataset; the preconditions for a second (an OK-VQA subset) were empirically shown not to hold — the 3B parameter hit rate on it is only 25.5% and the questions are all perceptual with no evidence field, making conflict conditions impossible to construct. Hence “the conclusions are not specific to this dataset’s format” is something we cannot prove, and this is the most substantial limitation of this paper.

4.6.2 Limitations of criteria and measurement

The main criterion is a string rule, not a semantic judgement; complex or non-English outputs may be missed, which creates no false positives but underestimates the effect. The absolute magnitude for `ctx.hedge` is untrustworthy, and on the Qwen2.5 family the two criteria give opposite signs (alias -4.2 pp, $n = 24$; `negation-aware` $+11.6$ pp, $n = 584$). The former misses the $n \geq 50$ threshold and by rule takes part in no sign comparison, but the disagreement must be disclosed: it is a counterexample to “criterion choice can move the sign” and it occurs on one of our own main-result cells — one that carries none of the paper’s claims, since removing it leaves the sign sequence unchanged and the span drops only from 56.5 to 55.3 pp (4.1.3). Position preference makes some forced-choice conditions unusable (4.2.3), and the same risk may exist in other forced-choice results without order control. The eighth stance’s sign (4.7) holds only under the `negation-aware` criterion and its magnitude is sensitive to prompt length; we claim only the sign.

4.6.3 Limitations of the mechanistic explanation and of priority of discovery

The mechanistic explanation is a behavioural-level inference, not a representational probe: we explain what behaviour arises under what conditions, but not how the model internally represents the conflict, which would require activation access and representational probes we have not done. That “instruction stance explains the disagreement between two published works” is an external inference; we lack their original internal records and cannot verify it. Nor do we claim priority for either sign direction or for originating “the effect sign can cross zero” itself (2.2); what we claim is placing them on the same pre-registered authorisation ladder and obtaining all signs on the same batch of samples.

Figure 11. Each answer matches both the parametric truth and the context value, so the alias-only criterion judges BOTH; the negation-aware criterion recognises explicit rejections of the form `isnot` or `incorrectlystates` and re-judges PARAM. All four come from the same cell but lie outside the fixed-seed sample of Appendix A.8; they were picked by hand for how typical their rejection phrasing is, so they are illustrative only and enter no count.



both criteria match two values -> alias-only marks BOTH; negation-aware detects the rejection cue -> PARAM

Fig. 11 Four verbatim answers discarded by the alias-only criterion and rescued by the negation-aware one.

4.7 The eighth stance adjudicate: a pre-registered prediction test

4.7.1 Why a prediction test is needed

The preceding sections give descriptive findings: the seven stances' signs occupy the same positions in both families, the first two negative and the last five positive. A descriptive finding cannot be distinguished from "these signs were these numbers all along" — as long as the stances are fixed after seeing the results, any sign structure can be narrated as a regularity. We therefore performed a falsifiable test before collecting the data: treat "signs are ordered with authorization strength" as the hypothesis, use the first six stances to fix a sign rule on the training family, and predict the sign of the eighth stance **adjudicate**, never measured in this work, with the prediction, decision rule, and numeric interval all fixed in writing before seeing any output of this stance (**prereg_stance8.md**). Predicting unseen data guards against selective reporting [28, 40]. The eighth stance sits deliberately between **own_only** and **prior**: it assigns no side priority but asks the model to judge the reliability of the context and of its own knowledge, testing the authorization order itself rather than the wording of any one stance.

4.7.2 The pre-registered prediction and decision rule

The sign rule is taken from the first six stances: those preceding **neutral** in the authorization order are predicted negative, the rest positive. The order (weak to strong) is **ctx_only** | **orig** | **neutral** | **adjudicate** | **own_only** | **len_ctrl** | **prior**; **ctx_hedge**

does not enter it, its sign being unclaimable in the Qwen2.5 family (the two criteria give opposite signs and the alias denominator misses the $n \geq 50$ threshold), so it is retained as an outlier diagnostic arm. The decision rule was fixed before seeing the data: both families positive is a successful prediction, both negative a failure requiring the core claim to be downgraded, and one of each a partial failure; when the sign is positive but the magnitude falls outside the predicted interval, the two are reported separately.

4.7.3 Measured results

The anchoring set follows D^* of the main experiment, $|D^*| = 649$, and is not recomputed (the eighth stance does not change the anchoring protocol); the criterion is the **negation**-aware one, as elsewhere. Table 12 gives the four arms side by side under the two criteria; the decision and claimability vary with the criterion, so the two must be read together (4.7.5 explains the degradation under the alias criterion).

Table 12 Measured results of the prediction test for the eighth stance under the two criteria

Family	Scale pair	Arm	Criterion	Δ (pp)	p	n	Claim
Qwen3-VL	2B $\rightarrow$ 8B	adjudicate	neg.-aware	+46.1	2.1×10^{-50}	360	Yes
Qwen3-VL	2B $\rightarrow$ 8B	adjudicate _len	neg.-aware	+45.6	1.1×10^{-47}	344	Yes
Qwen2.5-VL	3B $\rightarrow$ 7B	adjudicate	neg.-aware	+15.7	9.2×10^{-15}	485	Yes
Qwen2.5-VL	3B $\rightarrow$ 7B	adjudicate _len	neg.-aware	+29.0	4.1×10^{-33}	465	Yes
Qwen3-VL	2B $\rightarrow$ 8B	adjudicate	pure alias	+2.4	1.00	41	No, $n < 50$
Qwen3-VL	2B $\rightarrow$ 8B	adjudicate _len	pure alias	+0.0	1.00	29	No, $n < 50$
Qwen2.5-VL	3B $\rightarrow$ 7B	adjudicate	pure alias	+2.3	0.54	175	No, $p > 0.05$
Qwen2.5-VL	3B $\rightarrow$ 7B	adjudicate _len	pure alias	+8.0	2.3×10^{-2}	150	Weak

The sign prediction holds: in both families Δ is positive and highly significant, so by the pre-registered rule the core claim for this stance is upgraded from descriptive to a predictable regularity — the authorization order predicts the sign of a previously unmeasured stance. The magnitude prediction fails: the pre-registered intervals are $[+12, +28]$ for the Qwen3 family and $[+22, +34]$ for the Qwen2.5 family, and both measured values fall outside, in opposite directions (+46.1 above the upper bound, +15.7 below the lower). The hypothesis that **adjudicate** should fall between **own_only** and **prior** does not hold: the Qwen3 value exceeds that family’s highest existing stance, **prior** (+28.0).

The two families fall at the two ends of the ladder rather than in the same segment, the most substantial result of this section. It sharpens the cross-family heterogeneity of 4.5.1: the same stance differs between families not merely in magnitude but in its position relative to the existing stances. The authorization order can therefore predict the sign, but not the relative position of the sign across families.

No saturation artifact is observed: across the five scale points the parameter-knowledge proportion for `adjudicate` is 0.382, 0.512, 0.550 (Qwen3-VL 2B/4B/8B) and 0.493, 0.683 (Qwen2.5-VL 3B/7B), rising monotonically with scale, so +46.1 is an ordered effect rather than a ceiling artifact. The paired flips in the Qwen3 family are one-directional ($\uparrow 0$, $\downarrow 166$).

4.7.4 The length-control arm

`adjudicate.len` appends "Take your time to think." after the same prompt, growing the instruction from 37 to 42 words; its pre-registered purpose is to rule out length as an alternative explanation. The two families have the same sign, so length does not drive the sign — supporting the main conclusion — but it clearly drives the magnitude, and in opposite ways: the Qwen2.5 family moves from +15.7 to +29.0 (+13.4 pp, inside the predicted interval) while the Qwen3 family barely moves (−0.5 pp). Differing by one sentence of prompt, the two families yield results nearly 14 pp apart, showing that the magnitude for this stance is sensitive to prompt length — itself another instance of a measurement protocol affecting conclusions.

4.7.5 The same test under the alias criterion

Under the rule of 4.4 the same test is recomputed under the alias criterion, in the lower half of Table 12 (the last four rows); this reading is clearly weaker, and we list it faithfully rather than hiding it. The two Qwen3 cells' decidable sample counts are 41 and 29, both below the $n \geq 50$ threshold, so by 4.4 they take part in no sign decision; the Qwen2.5 family's main arm has $n = 175$, passing the threshold, but $p = 0.54$, likewise constituting no claim. The collapse has the same cause as in 4.1.1: the alias criterion judges an explicit rejection naming both values at once as undecidable, and `adjudicate` precisely induces such rejections (the alias decision rate of Qwen3 8B is only 0.073 against 0.894 under `negation-aware`). The sign claim therefore holds only under the `negation-aware` criterion; we take that as the reporting protocol because the criterion choice was fixed under 4.4 before seeing this stance's data.

4.8 The common structure of the three measurement biases

The three biases are instances of one class of failure. In the criterion defect (4.1.1), alias matching judges an explicit rejection that names two values at once as undecidable and removes it, yet such answers are the most explicit class of following parametric knowledge, and the removed amount rises with scale. In the denominator collapse (4.4.2), when paired decidable samples fall to single digits Δ can only take a degenerate value, yet in the table it looks like the strongest cell. In the position-preference self-refutation (4.2.3), rerunning with the option contents swapped leaves the output almost unchanged while the letter preference flips, showing the quantity measured is

not a content judgement. What the three share is that the quantity measured and the quantity reported are not the same, with the bias direction systematic in each; the two corrections (the **negation**-aware criterion and position control) both point the same way — first confirm what is being measured, then report the number measured. Most numbers in knowledge-conflict evaluation come from a default protocol, and several of its internal decisions — how explicit rejections are counted, how far the denominator collapses, how the options are ordered — can each move the sign.

5 Conclusion

Under controlled counterfactual edits and an anchoring protocol, we turn ”does model scale affect resistance to tampered retrieved evidence” into a decidable experiment: holding the conflict construction, decision criterion, models, and samples fixed, we vary only the system prompt’s authorisation strength for parametric knowledge. The sign of this effect changes in an orderly way with authorisation strength and crosses zero in two independent model families, with large models adhering more strongly to parametric knowledge under strong authorisation (+28.0 and +33.7 pp) and more inclined to follow the context under weak authorisation (−27.3 and −8.4 pp).

The three methodological results supporting this conclusion are equally binding: explicit dismissal judged as undecidable systematically underestimates large models; ratio-type statistics lose meaning when the denominator collapses to single digits; and high accuracy under a forced-choice format can be a product of position preference. None disappears when the criterion or template is changed, and each at some point reversed the direction of one of our own conclusions.

We further turn the sign structure into a falsifiable test: before collecting the eighth stance level we fix the sign rule from the first six and pre-register a prediction for the eighth. The sign prediction holds while the magnitude prediction fails, and the two families fail in opposite directions—Qwen3-VL’s +46.1 pp exceeds the family’s highest stance, and Qwen2.5-VL’s +15.7 pp falls below the lowest anchor. The authorisation order can therefore predict the sign of an unmeasured stance but not its relative position across families. Under the alias criterion the decidable samples for this level collapse to 41 and 29, below our self-imposed threshold, so this cell holds only under the **negation**-aware criterion—criterion choice here is the precondition for whether the level can be detected at all.

The magnitude is not monotone with scale: the same quantity can appear as a continuous climb or a discontinuous jump depending on difficulty and scoring convention [26, 27]. We do not claim priority for either the negative or the positive sign. What we claim is evidence for taking authorisation direction as a class of manipulation not previously measured independently, together with these three directionally clear measurement biases.

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